Use Cases

for

URECAMatch

Version 1.0

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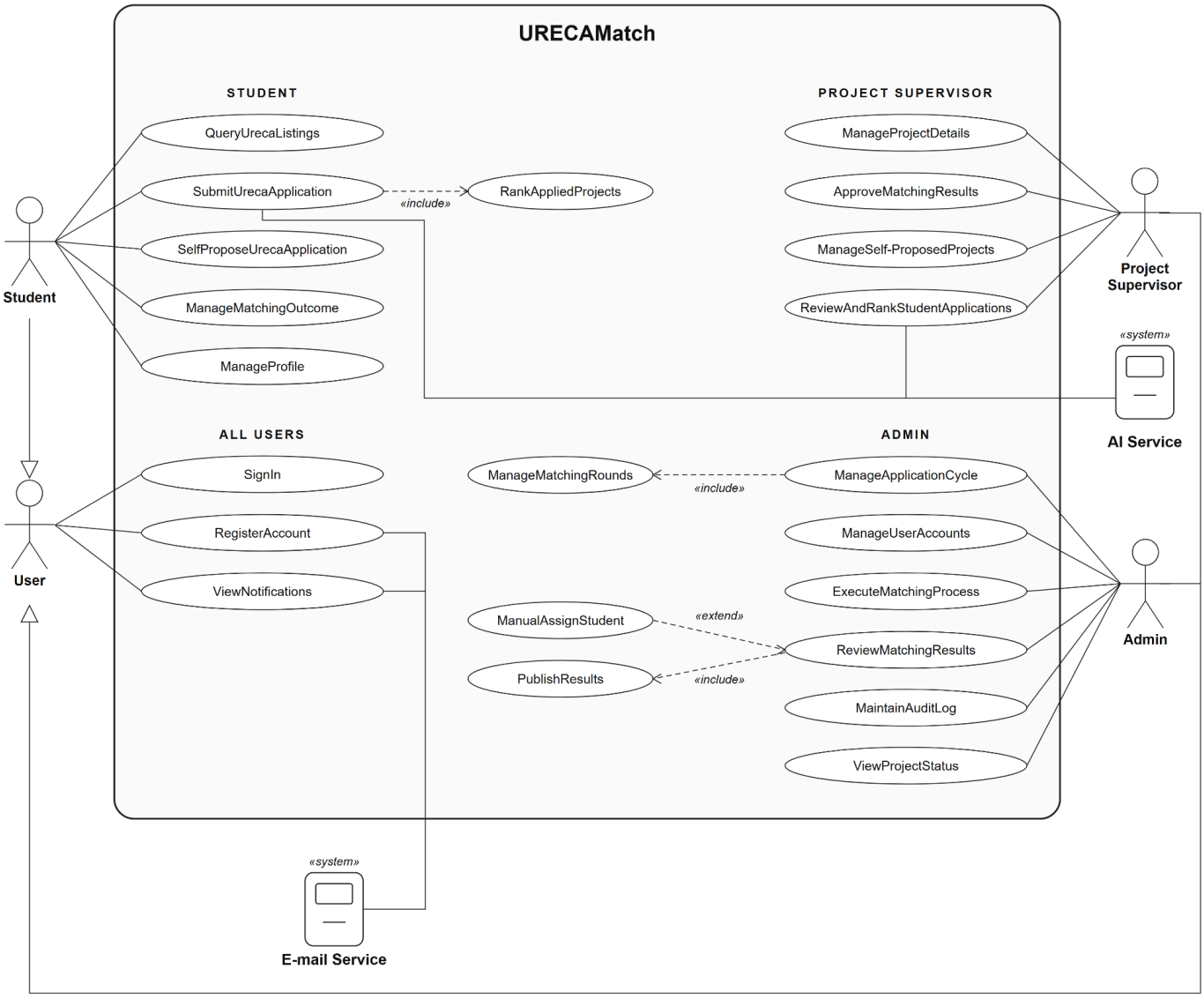
SCEF GRP 4

8/11/2026

Revision History

Name	Date	Reason For Changes	Version
Ian Tan	8/11/2026	LAB 3 and LAB 4: use cases aligned with SRS Section 4 and the final application	1.0

1. Use Case Diagram



2. Use Case Description

2.1 For Functional Requirement #1 (Admin)

2.1.1 ManageApplicationCycle

Use Case ID:	#1-1		
Use Case Name:	ManageApplicationCycle		
Created By:	Nicholas Ng	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin creates a new Application Cycle or modifies an existing one. An Application Cycle consists of a global start/end date, a pre-application viewing period (0 to 60 days, where project listings are visible but applications are disabled), and up to three sequential matching rounds (each comprising an Application Window, a Review Window and an Approval Window, with lengths set by the Admin).
Preconditions:	The Admin is signed in.
Postconditions:	The overall Application Cycle and its constituent rounds are successfully created or updated in URECAMatch. When a new cycle is created, the previous cycle moves to the history, its published projects are closed, unanswered offers become Timed out and unanswered proposals become Withdrawn.

Priority:	High
Frequency of Use:	Several times per year.
Flow of Events:	1. The Admin navigates to the Application Cycle page and selects either "New cycle" or "Cycle settings" for an existing cycle. 2. The System displays the Application Cycle configuration. 3. The Admin configures the global cycle parameters, including the overall start date, end date, and the length of the pre-application viewing period (0 to 60 days). 4. The Admin configures the individual matching rounds (ManageMatchingRounds). 5. The Admin clicks "Create cycle" (new cycle) or "Save settings" (Cycle settings) to submit the configuration. 6. The System validates the provided parameters. 7. The System saves the Application Cycle and updates the active records.
Alternative Flows:	AF-1: The Admin clicks "Cancel" 1. The System discards filled fields or changes and exits to the Application Cycle page.

Exceptions:	E1: The closing date of an Application Cycle is not after the opening date. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E2: The Admin attempts to create a new Application Cycle with dates that overlap an existing Application Cycle. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E3: The Admin attempts to create a new Application Cycle with a closing date that has already lapsed. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page. E4: The Admin made no changes to the Application Cycle. 1. The System greys out the "Save settings" button and disables its interactivity. E5: The Admin attempts to create a new Application Cycle while the current cycle's closing date has not passed. 1. The System rejects this change. 2. The System displays the rejection reason to the Admin, and remains on the configuration page.
Includes:	ManageMatchingRounds
Special Requirements:	If the closing date of an Application Cycle is modified during its active period, the System notifies all Students and Project Supervisors (REQ-CYCLE-3).

Assumptions:	Only one Application Cycle shall be active at any given time.
Notes and Issues:	-

2.1.2 ManageMatchingRounds

Use Case ID:	#1-2		
Use Case Name:	ManageMatchingRounds		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	6th September 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin configures up to three sequential matching rounds within the active application cycle. Each round consists of an Application Window (1 to 14 days), a Review Window (1 to 7 days) and an Approval Window (1 to 7 days, default 2), with lengths set by the Admin.
Preconditions:	The Admin is currently executing the ManageApplicationCycle use case.
Postconditions:	All matching rounds are valid and match to the whole application cycle.
Priority:	High
Frequency of Use:	Several times per year.

Flow of Events:	1. In "Cycle settings", the Admin sets the number of matching rounds (1 to 3). 2. The System generates the settings for up to three sequential matching rounds. 3. For each round, the Admin sets the Application Window length (1 to 14 days), the Review Window length (1 to 7 days), the Approval Window length (1 to 7 days) and, optionally, a start date. 4. The Admin chooses whether the applications of unmatched eligible Students roll over into the next round. 5. The System calculates the window dates of each round from its start date and window lengths. 6. The Admin clicks "Save settings" to confirm the round configuration. 7. The System validates that none of the rounds overlap with each other and that all rounds fall within the global Application Cycle dates.
Alternative Flows:	AF-1: Admin configures fewer than three rounds 1. The Admin sets the number of rounds to 1 or 2. 2. The System configures only those rounds for the cycle.
Exceptions:	E1: The calculated dates for a matching round overlap with another matching round. 1. The System rejects the configuration and highlights the overlapping dates. E2: The matching round dates fall outside the global Application Cycle start and end dates.

	1. The System rejects the configuration and prompts the Admin to adjust the round dates or the global cycle dates.
Includes:	-
Special Requirements:	A Matching Round can be changed or removed only before its Application Window starts. If the dates of a round change, the System notifies all Students and Project Supervisors of the new dates (REQ-ROUND-4).
Assumptions:	Every matching round follows the same rules, but each may have its own window lengths and start date.
Notes and Issues:	-

2.1.3 ExecuteMatchingProcess

Use Case ID:	#1-3		
Use Case Name:	ExecuteMatchingProcess		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin (monitors; the System runs the process automatically)
Description:	The System runs the stable-matching algorithm automatically when a round's Review Window ends, to assign students to projects based on ranked preferences and supervisor rankings (REQ-EXEC-6). The Admin

	monitors the process and its result. No Admin action is needed to start it.
Preconditions:	The current matching round's Review Window has ended.
Postconditions:	A proposed allocation is generated where no project exceeds available capacity and each student receives at most one match. Open slots carry over to the next round, and if the Admin enabled rollover, the applications of unmatched eligible Students carry over too, when the results are published.
Priority:	High
Frequency of Use:	Automatic, once per matching round.
Flow of Events:	1. When a round's Review Window closes, the System starts the matching process by itself. 2. The System verifies Student eligibility, excluding any Student who already holds a match this cycle or who declined or let an offer time out in an earlier round. 3. The System runs student-proposing deferred acceptance over the Students' preference rankings and the Supervisors' applicant rankings, within each listing's open slots. A Student is never matched to a listing whose Supervisor left them unranked. 4. If there is a subsequent round and rollover is enabled, the System marks the applications of unmatched eligible Students to carry over when the results are published. 5. The System commits match results to storage as Proposed and logs an immutable entry under the actor "System" with a timestamp. 6. The System notifies each Project Supervisor who has a proposed match.

Alternative Flows:	-
Exceptions:	E1: The round's Review Window has not ended. 1. The System does not start the matching process (REQ-EXEC-1). E2: The process has already run for the current round. 1. The System does not run it again, to prevent duplicate match generation (REQ-EXEC-2).
Includes:	-
Special Requirements:	-
Assumptions:	All submitted rankings are final upon deadline lapse.
Notes and Issues:	The Admin sees the result on the dashboard and on the round's matches page (ReviewMatchingResults).

2.1.4 ReviewMatchingResults

Use Case ID:	#1-4		
Use Case Name:	ReviewMatchingResults		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin

Description:	Admin reviews a round's matches, the declines made by Project Supervisors and the Students who were not matched. The page is read only. Reassignment is done on the Overrides page (ManualAssignStudent) and publication is automatic (PublishResults).
Preconditions:	1. The Admin is signed in. 2. The automated matching process has been executed. 3. Proposed matches exist in the system.
Postconditions:	The Admin has seen the round's matches with their status, the reasons for declines, and the unmatched Students.
Priority:	High
Frequency of Use:	Once per matching round
Flow of Events:	1. The Admin navigates to the "Application Cycle" page and opens a round's matches. 2. The System displays a summary (matched, rejected by supervisors, unmatched), the round's matches with their status and, for a decline, its reason, and the Students who applied but were not matched. 3. The Admin filters the matches by status. 4. For an unmatched Student the Admin selects "Manual Assign", which opens the Overrides page for that Student (ManualAssignStudent). 5. When the round's Approval Window ends, the System publishes the results by itself (PublishResults). 6. Students respond to their offers by the end of the next Application Window that has not closed, or by the cycle's closing date after the

	last round, after which an unanswered offer automatically becomes "Timed out" (REQ-PUB-3). 7. Unmatched students will wait for the next round's matching.
Alternative Flows:	AF-S3: Manual Reassignment of Student (done on the Overrides page) 1. The Admin selects "Reassign" next to a match that has not yet been offered to the Student. 2. The System displays available alternative projects with remaining slots. 3. The Admin selects an alternate project listing, enters a justification of at least 10 characters, and confirms. 4. The System decrements available slots on the new project and increments slots on the original project. 5. The System logs the previous allocation, new allocation, initiating Admin, and timestamp to the audit log. 6. If the round is already published, the System offers the reassigned match to the Student at once (REQ-REV-2).
Exceptions:	-
Includes:	-
Extends:	None (ManualAssignStudent extends this use case). PublishResults is included (flow step 5).
Special Requirements:	Reassignment operations shall be atomic to prevent slot over-allocation.

Assumptions:	-
Notes and Issues:	Manual overrides are recorded in the audit log.

2.1.5 PublishResults

Use Case ID:	#1-5		
Use Case Name:	PublishResults		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin (monitors; the System publishes automatically)
Description:	The System publishes a round's matching results automatically when its Approval Window ends, so that students and project supervisors can view their final allocations. Every match that was not declined becomes an offer (REQ-PUB-4). The Admin monitors the publication. No Admin action is needed to start it.
Preconditions:	1. The matching process has been executed for the current round. 2. The round's Approval Window has ended.
Postconditions:	Matching outcomes are transitioned to the published state, and automated notifications are sent to affected users.
Priority:	High

Frequency of Use:	Automatic, at the conclusion of each matching round.
Flow of Events:	1. When a round's Approval Window ends, the System publishes the round's results by itself. 2. The System transitions the matching results to the published state and sets the expiry of each offer. 3. The System triggers automated notifications to students and project supervisors. 4. If rollover is enabled, the System carries the applications of unmatched eligible Students into the next round.
Alternative Flows:	-
Exceptions:	E1: Matching has not run, or the Approval Window has not ended. 1. The System does not publish the results (REQ-PUB-1).
Includes:	-
Extends:	-
Special Requirements:	A round's results are published only once (REQ-PUB-1).
Assumptions:	The matching round data is fully recorded and ready for release.
Notes and Issues:	-

2.1.6 ManualAssignStudent

Use Case ID:	#1-6

Use Case Name:	ManualAssignStudent		
Created By:	Nicholas Ng	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin manually assigns a Student to a project. This function shall be reserved for when exceptional circumstances mandate a manual override of URECA Project assignments.
Preconditions:	The Admin is signed in.
Postconditions:	Students are manually assigned to a URECA Project.
Priority:	High
Frequency of Use:	Low (A few times per Application Cycle)
Flow of Events:	1. The Admin opens the Overrides page and finds the Student. 2. The Admin clicks "Manual Assign" for a Student with no project, "Reassign" for a Student whose round has not published, or "Manual Unassign". 3. The Admin selects a published URECA Project with an open slot and enters a justification of at least 10 characters. 4. The Admin clicks "Confirm". 5. The System assigns the Student to the current round and records the assignment, justification, initiating Admin and timestamp in

	the audit log. The Student receives the offer with that round's results, or at once if those results are already published (REQ-ASSIGN-5).
Alternative Flows:	AF-1: Unassign Student. 1. The Admin selects "Manual Unassign" next to an assigned Student and enters a reason of at least 10 characters. 2. The System frees the slot, notifies the Project Supervisor (and the Student, if they had been offered the place), and records the action in the audit log.
Exceptions:	E1: Student is already assigned to a URECA Project 1. The System displays a warning message, and the URECA Project is not assigned. 2. Admin unassigns the current URECA Project from the Student (AF-1). 3. Admin repeats the ManualAssignStudent use case from Step 2.
Includes:	-
Extends:	- ReviewMatchingResults
Special Requirements:	-
Assumptions:	-
Notes and Issues:	The Overrides page lists every active student in the cycle with where they stand (accepted, offer sent, matched, or no project).

2.1.7 ManageUserAccounts

Use Case ID:	#1-7		
Use Case Name:	ManageUserAccounts		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin creates, modifies, or deactivates user accounts in the system
Preconditions:	The Admin is signed in.
Postconditions:	The user's account details, assigned roles, or activation status are updated in persistent storage.
Priority:	High
Frequency of Use:	Occasionally, mostly at the start of a cycle
Flow of Events:	1. The Admin navigates to the "User Management" page. 2. The System displays the full directory of users with role and status filters. 3. The Admin selects a specific user account to edit. 4. The Admin updates the user's details or role (Student, Project Supervisor, Admin).

	5. The System asks the Admin to confirm the role change (REQ-USER-5). 6. The Admin clicks "Save". 7. The System commits the changes and logs the action.
Alternative Flows:	AF-S1: Create User Account: 1. The Admin selects "New user". 2. The Admin enters the name, NTU e-mail, password and role. 3. The System creates the account and logs the action (REQ-USER-1). AF-S3: Deactivate User Account: 1. The Admin selects "Deactivate" next to an active user. 2. The System prompts for confirmation. If the change would close active applications, listings, proposals or matches, the System lists them and requires the Admin to acknowledge the warning (REQ-USER-5). 3. The Admin confirms, and the System marks the user account as deactivated and ends the user's active sessions.
Exceptions:	E1: Creation of Admin role by unauthorized user: 1. The System rejects the role change if the current session does not hold Admin privileges. E2: The Admin tries to change their own role, deactivate their own account, or leave the system without an active Admin: 1. The System refuses the change (REQ-USER-4).

Includes:	-
Special Requirements:	Admin account creation and role elevation operations shall be strictly restricted to existing authenticated Admins.
Assumptions:	All registered user accounts have valid NTU domain emails.
Notes and Issues:	Deactivated users shall have their active sessions immediately terminated.

2.1.8 MaintainAuditLog

Use Case ID:	#1-8		
Use Case Name:	MaintainAuditLog		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin views immutable audit records of all administrative operations: cycle and round changes, matching runs, manual assignments and reassignments, publication of results, and account changes.
Preconditions:	The Admin is signed in.
Postconditions:	Audit log entries are retrieved and rendered in read-only mode

Priority:	Medium
Frequency of Use:	After matching cycles or as needed for review
Flow of Events:	1. The Admin navigates to the "Audit Log" page. 2. The System displays timestamped records of all administrative operations along with the initiating Admin, or the System for actions it takes by itself, newest first, 50 at a time with "Load more" (REQ-AUDIT-3, REQ-AUDIT-4). 3. The Admin filters or inspects specific log entries. Search and filters cover the whole log, not only the loaded entries.
Alternative Flows:	-
Exceptions:	E1: Modification attempt: 1. The System strictly prevents any edit or delete operations on audit records.
Includes:	-
Special Requirements:	Audit log entries shall be write-once, append-only, and completely immutable to all users including Admins.
Assumptions:	Every administrative operation listed in REQ-AUDIT-1 automatically triggers an audit log write.
Notes and Issues:	Log entries are kept after a cycle closes (SRS 5.3.4 and 6.1).

2.1.9 ViewProjectStatus

Use Case ID:	#1-9

Use Case Name:	ViewProjectStatus		
Created By:	Robin Teoh	**Last Updated By:**	Robin Teoh
Date Created:	1st November 2026	**Date Last Updated:**	1st November 2026

Actor:	Admin
Description:	Admin views every project of the active cycle and the students each project holds, read only (REQ-PSTAT-1, REQ-PSTAT-2).
Preconditions:	The Admin is signed in.
Postconditions:	-
Priority:	Medium
Frequency of Use:	Several times per matching round.
Flow of Events:	1. The Admin selects "Project Status" in the sidebar. 2. The System displays every project with its supervisor, school, slots filled, applicant count and status, with a search box and School, Supervisor and Status filters. 3. The Admin selects a project. 4. The System displays the project's details and the students it holds, each with their status and round.

Alternative Flows:	AF-1: The Admin searches or filters the list. 1. The Admin types in the search box or chooses a School, Supervisor or Status filter. 2. The System returns the projects that match all the criteria. AF-2: The Admin opens a student from a project. 1. The Admin selects the student's name on the project page. 2. The System opens the Overrides page filtered to that student (ManualAssignStudent).
Exceptions:	E1: The list cannot be retrieved. 1. The System displays an error message.
Includes:	-
Special Requirements:	The page is read only.
Assumptions:	-
Notes and Issues:	The students shown on each project page are described in REQ-ASSIGN-6.

2.2 For Functional Requirement #2 (Student)

2.2.1 QueryUrecaListings

Use Case ID:	#2-1
Use Case Name:	QueryUrecaListings

Created By:	Nicholas Ng	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student
Description:	Student browses and queries available URECA Project Listings using keyword search and/or filters, and sees the projects that best fit their research interests.
Preconditions:	1. The Student is signed in. 2. A pre-application period OR an active matching round is ongoing.
Postconditions:	-
Priority:	High
Frequency of Use:	Very High
Flow of Events:	1. The Student selects "Browse projects". 2. The System displays the published URECA Project Listings, each with its title, offering school, Project Supervisor, open slots, number of applicants and, when demand is high, a High demand label. 3. If the Student has research interests, may still apply this cycle, and an application window is open, the System also shows up to six recommended projects above the listings, each with a match score from 0 to 100 and the reasons for it.

	4. The Student enters a keyword in the search field. 5. The System returns listings that match the query.
Alternative Flows:	AF-S3: The Student opens a recommended project. 1. The Student selects a recommended project. 2. The System displays the full details of that project. AF-S4: The Student uses additional filters to query the listings. 1. The Student chooses to filter by school, Project Supervisor name, or topic tag. 2. The System returns listings that match all search/filter criteria. AF-S5: The Student sorts the listings by best match. 1. The Student chooses "Best match". 2. The System orders the listings by match score, highest first. AF-S6: The Student chooses to return to an unfiltered view of the listings. 1. The Student clicks "Clear". 2. The System returns to the default, unfiltered view of all URECA Project Listings. AF-S7: The Student asks how a match score is made. 1. The Student selects "How it works". 2. The System explains the three parts of the score: shared topics (up to 50), prerequisites passed (up to 30) and applicants per open slot (up to 20).

Exceptions:	E1: The search request from the server times out. 1. The System informs the Student accordingly via an error message. E2: The Student has no research interests. 1. The System shows no recommendations and offers a link to add research interests on My profile. E3: The Student already holds a project, or has declined or not answered a match offer. 1. The System shows no recommendations. The System tells the Student, kindly, that they declined the offer or let it expire and are out of the remaining rounds this cycle, and that they can still browse. The listings remain viewable, shown faded with a lock marked "Not eligible", but the Student cannot apply. E4: No application window is open yet (no cycle, or the pre-application period). 1. The System tells the Student they are early and when applications open, and that they can browse and shortlist projects now. No projects are recommended, and the listings are shown faded with a lock until applications open. E5: The cycle is between application windows, or has closed. 1. The System reports where the cycle is: the round is under review, the System is waiting for its results, the results are out (the Student's offer, or that the Student was not matched and what carries over to the next round), or the cycle has closed. No projects are recommended, and the listings are shown faded with a lock.
Includes:	-

Special Requirements:	Each query request shall complete within 5 seconds. Match scores are shown only to the Student and are never shown to Project Supervisors (REQ-BROWSE-8).
Assumptions:	There will be ample URECA Project Listings posted by Project Supervisors during an active Application Cycle. Topic tags and research interests are chosen from the same standard topic list, so they can be compared exactly.
Notes and Issues:	Recommendations are calculated by the System from the score in REQ-BROWSE-6.

2.2.2 SubmitUrecaApplication

Use Case ID:	#2-2		
Use Case Name:	SubmitUrecaApplication		
Created By:	Nicholas Ng	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student
Description:	Student applies for a URECA Project by submitting an Application to the desired URECA Project Listing.

Preconditions:	1. The Student is signed in. 2. The current round's Application Window is open. 3. At least one URECA Project Listing is available for application. 4. Student has completed their profile. 5. Student does not already hold a project this cycle and has not declined or missed a match offer.
Postconditions:	An Application for the desired URECA Project has been submitted to a Project Supervisor for consideration.
Priority:	High
Frequency of Use:	Medium (a few times per Student)
Flow of Events:	1. The Student views the full details of a URECA Project Listing. 2. The Student clicks "Apply". 3. The Student writes a Statement of Interest of at least 50 characters and may attach their uploaded supporting documents. 4. Optionally, the Student clicks "Check my application". The System shows tips on the statement and the attached documents. It gives no score and saves nothing (REQ-APPLY-6 to REQ-APPLY-8). 5. The Student clicks "Submit application". 6. The System validates the submitted inputs and documents. 7. The System submits the application for Project Supervisor consideration. The application

	becomes the Student's lowest preference (RankAppliedProjects).
Alternative Flows:	AF-S3: Student clicks "Back to project". 1. The System displays a confirmation prompt about unsaved Application data. 2. Student clicks "Confirm". 3. The System returns to displaying the full descriptive view of the selected URECA Project.
Exceptions:	E1: Student leaves the Statement of Interest blank or under 50 characters. 1. The System displays the error message to the Student upon clicking "Submit application", and remains on the Application page. E2: Student uploads documents that are not in PDF form. 1. The System rejects the upload while displaying the appropriate error feedback to the Student. E3: The round's Application Window ends while Students are editing their application before submission. 1. The Student clicks "Submit application". 2. The System rejects the submission and displays the reason for rejection. 3. The System returns to displaying the full details of the selected URECA Project, while greying and disabling interactivity of the "Apply" button. E4: Student already holds a project this cycle, or has declined or missed a match offer. 1. The System shows the project as "Ineligible" and does not offer the "Apply" button. E5: The Student runs more than 6 checks in 10 minutes.

	1. The System refuses the extra check and asks the Student to wait (REQ-APPLY-9).
Includes:	RankAppliedProjects
Special Requirements:	The Statement of Interest shall be at least 50 characters long. The System rejects an application if the Student's profile is incomplete, the Student is ineligible or already holds a project this cycle, the listing is not published or has no open slots, or the Student already has an active application to that listing (REQ-APPLY-3). A Student who holds an open match offer cannot apply (REQ-APPLY-5).
Assumptions:	-
Notes and Issues:	-

2.2.3 RankAppliedProjects

Use Case ID:	#2-3		
Use Case Name:	RankAppliedProjects		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student
Description:	This feature allows a student to arrange and order their submitted URECA project applications in order of preference during the active application window.

Preconditions:	1. The Student has submitted at least one URECA project application. 2. The current round's Application Window is open.
Postconditions:	The student's priority rankings for their submitted applications are successfully recorded and saved in the system.
Priority:	High
Frequency of Use:	Medium (a few times per Student)
Flow of Events:	1. The Student navigates to "My applications". 2. The System displays the list of currently submitted URECA project applications. 3. The Student moves an application up or down to change its place in the preference order. 4. The System validates that every active application appears exactly once, with sequential preference numbers. 5. The System saves the new order immediately, with no separate save button, and shows "Order saved".
Alternative Flows:	AF-1: Student withdraws an application that is still under review. 1. The Student clicks "Withdraw application" and confirms. 2. The System withdraws the application and renumbers the remaining preference ranks (REQ-APPLY-4).

Exceptions:	E1: Student attempts to change the ranking outside of the active application window. 1. The System displays an error message notifying the Student that the window is closed, and prevents the update. E2: The ranking does not contain every active application exactly once. 1. The System rejects the update and keeps the previous order (REQ-SRANK-2).
Includes:	-
Special Requirements:	Each active application appears exactly once in the ranking. The ranking cannot be changed outside the active application window.
Assumptions:	The student has active project applications to rank.
Notes and Issues:	-

2.2.4 SelfProposeUrecaApplication

Use Case ID:	#2-4		
Use Case Name:	SelfProposeUrecaApplication		
Created By:	Nicholas Ng	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student

Description:	Students self-propose a URECA Project and submit it to their desired Project Supervisor for approval, and may withdraw it while the Supervisor has not answered.
Preconditions:	1. The Student is signed in. 2. The current round's Application Window is open. 3. The Student does not already hold a project this cycle and has not declined or missed a match offer.
Postconditions:	A self-proposed URECA Application has been submitted for Project Supervisor consideration, and the Supervisor has been notified.
Priority:	High
Frequency of Use:	Low (Most Students will apply to listed URECA Projects)
Flow of Events:	1. Student clicks on "Propose a project". 2. The System displays the Application page for a Self-Proposed URECA Project, in four steps: Supervisor, Your project, Why this supervisor, and Review and send. 3. The Student fills in their proposed URECA Project title, write-up and statement, choose research interests from the standard topic list, attach supporting documents from their profile, and choose a Project Supervisor. 4. The Student clicks "Send proposal". 5. The System displays a confirmation dialog stating that, once accepted, the proposal cannot be changed and all the Student's other applications and proposals are withdrawn. 6. The Student clicks "Send proposal" to confirm.

	7. The System validates all required fields, submits the proposal for Project Supervisor consideration, and notifies the Supervisor. 8. The System displays a confirmation page naming the Supervisor, with links to browse projects and to track the proposal.
Alternative Flows:	AF-S3: Student clicks on the "Browse projects" tab without submitting the Application. 1. The System displays a confirmation prompt about unsaved Application data. 2. Student clicks "Confirm". 3. The System returns to displaying the URECA Project Listings. AF-S6: Student cancels at the confirmation dialog. 1. The System closes the dialog, keeps the entered data and sends nothing. AF-1: Student withdraws a proposal that the Supervisor has not answered. 1. On "My applications", the Student clicks "Withdraw" on the pending proposal. 2. The System asks the Student to confirm. 3. The Student clicks "Confirm". 4. The System marks the proposal as Withdrawn and notifies the Supervisor.

Exceptions:	E1: Student leaves one or more required field(s) blank or too short. 1. The System displays the error message to the Student upon clicking "Send proposal", and remains on the Application page. E2: The description is over 500 words or the statement is over 300 words. 1. The System rejects the proposal and names the field to shorten. E3: The Application Window ends while Students are editing their application before submission. 1. The Student clicks "Send proposal". 2. The System rejects the submission and displays the reason for rejection. 3. The System remains on the proposal page, while greying and disabling interactivity of the "Send proposal" button. E4: The Student already has a proposal waiting with the chosen Supervisor. 1. The System rejects the proposal and asks the Student to withdraw the earlier one or wait for the Supervisor's reply. E5: The Student already holds a project, or has declined or missed a match offer. 1. The System rejects the proposal and shows the page as view only. E6: The Student tries to withdraw a proposal the Supervisor has already accepted or declined. 1. The System refuses, as the decision is final.
Includes:	-

Special Requirements:	The title shall be 5 to 150 characters, the description at least 50 characters and at most 500 words, and the statement at least 50 characters and at most 300 words. At least one research interest is required (REQ-PROPOSE-1, REQ-PROPOSE-4 to 6).
Assumptions:	-
Notes and Issues:	A Student may send proposals to several Supervisors, with one pending proposal per Supervisor. When one is accepted, the others are withdrawn and their Supervisors notified.

2.2.5 ManageMatchingOutcome

Use Case ID:	#2-5		
Use Case Name:	ManageMatchingOutcome		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student
Description:	Students review the published matching allocation and decide whether to accept or decline the offered project slot.
Preconditions:	1. The Student is signed in. 2. Matching results have been published.

Postconditions:	Match status is recorded as accepted, declined or timed out. Accepting automatically closes out all other active applications and pending proposals.
Priority:	High
Frequency of Use:	Once per cycle.
Flow of Events:	1. The Student navigates to "My applications". 2. The System displays the match offer ("Match offer, Round N") with its respond-by date. 3. The Student reviews the project title and assigned Project Supervisor. 4. The Student clicks "Accept offer". 5. The System displays a warning modal stating that accepting will withdraw all other pending applications and proposals. 6. The Student selects "Accept" to confirm. 7. The System records the acceptance, fills one slot of the project, sets all other applications submitted by the Student to withdrawn, withdraws the Student's pending proposals and notifies the Supervisors of those proposals. If the accepted place was the project's last open slot, the System also withdraws every other Student's application to that project, renumbers their remaining preferences and notifies them (REQ-OUTCOME-6).

Alternative Flows:	AF-S4: Student declines match offer. 1. The Student clicks "Decline". 2. The System displays a confirmation dialog warning that the student will be ineligible for subsequent matching rounds. 3. The Student confirms. 4. The System marks the allocation status as declined. 5. The System flags the Student account as ineligible for any subsequent matching rounds in the current cycle. 6. The System withdraws the Student's pending proposals and notifies their Supervisors. AF-S5: The Student does not respond by the end of the next Application Window that has not closed, or by the cycle's closing date after the last round (REQ-PUB-3). 1. The System automatically transitions the match status to "Timed out". 2. The System flags the Student account as ineligible for subsequent rounds. 3. The System withdraws the Student's pending proposals and notifies their Supervisors.
Exceptions:	E1: Student attempts to access outcomes before publication. 1. The System shows current application submission status and hides the offer acceptance interface. E2: Student is already registered to a self-proposed project. 1. The System refuses the acceptance and explains that the Student already holds a project this cycle.

Includes:	ViewMatchingResult
Special Requirements:	Immediate in-app visual confirmation upon selection.
Assumptions:	Each student holds at most one open match offer at a time.
Notes and Issues:	Once confirmed, acceptance cannot be undone through the student UI.

2.2.6 ManageProfile

Use Case ID:	#2-6		
Use Case Name:	ManageProfile		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Student
Description:	Student completes, views, and updates personal academic details, completed modules, research interests, and transcript documents.
Preconditions:	The Student is signed in.
Postconditions:	Updated profile data and uploaded transcript are saved to the student record.
Priority:	High

Frequency of Use:	Upon first sign-in and occasionally updated.
Flow of Events:	1. The Student navigates to "My profile". 2. The System displays the student's school, degree, CGPA, year of study, completed modules, and research interests. A Student with no transcript sees only an upload area until one is uploaded. 3. The Student updates fields, choosing completed modules from the NTU course catalogue (searched by code or title) and research interests from the standard topic list, and uploads an updated academic transcript PDF. A transcript can be replaced but not deleted (REQ-PROF-8). 4. The Student clicks "Save changes". 5. The System validates the file and inputs, then saves the profile.
Alternative Flows:	AF-S3: The Student fills the form from the transcript. 1. The Student uploads a transcript PDF. 2. The System reads the CGPA, programme, year of study and passed modules, and fills them into the form for the Student to check. 3. Nothing is saved until the Student clicks "Save changes".

Exceptions:	E1: Invalid transcript format or file size > 5 MB. 1. The System rejects the upload and displays an error message. E2: The Student uploads more than 5 supporting documents. 1. The System rejects the upload and states the limit. E3: A module or research interest is not in the NTU course catalogue or the standard topic list. 1. The System rejects the change and names the field. E4: The file is not an NTU transcript (it does not state a Cumulative GPA and list courses passed). 1. The System does not store the file, leaves the profile unchanged and warns the Student (REQ-PROF-4 to REQ-PROF-6).
Includes:	UploadTranscript
Special Requirements:	File upload shall be restricted to PDF format with a maximum size limit of 5 MB. Completed modules and research interests may only be chosen from the NTU course catalogue and the standard topic list.
Assumptions:	First-time sign-in requires completion of the profile before browsing or applying for projects.
Notes and Issues:	-

2.3 For Functional Requirement #3 (Project Supervisor)

2.3.1 ManageProjectDetails

Use Case ID:	#3-1		
Use Case Name:	ManageProjectDetails		
Created By:	Robin Teoh	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	24th October 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to create, update, and close research project listings.
Preconditions:	1. Project Supervisor is logged in and authenticated. 2. Listings may be created before an Application Cycle opens or during it.
Postconditions:	The project listing is created, updated, or closed, and its current state is saved in the system.
Priority:	High
Frequency of Use:	Medium

Flow of Events:	1. The system displays the Project Supervisor's existing project listings under Published, Drafts and Closed, and prompts them to select CREATE, UPDATE, or CLOSE. 2. If the Project Supervisor selects CREATE, the system prompts for the project title, description, and prerequisites. Prerequisites are searched in the NTU course catalogue and shown with their course titles. 3. The Project Supervisor enters the number of available slots, the offering faculty and the research location. 4. The Project Supervisor chooses topic tags from the standard topic list. 5. The Project Supervisor may add a cover image and optional details: tasks and outputs, skills gained, hours per week, mode (on site, hybrid or remote), up to 4 photos, and one YouTube or Vimeo video link. 6. The Project Supervisor selects PUBLISH or SAVE AS DRAFT. 7. The system validates the entered data and saves the listing as published, or as a draft.

Alternative Flows:	AF-S1: If the Project Supervisor selects UPDATE. 1. The system displays the selected project's existing details for editing. 2. The Project Supervisor modifies the details and selects SAVE. 3. The system validates and saves the updated listing. AF-S2: If the Project Supervisor selects CLOSE. 1. The system prompts the Project Supervisor to confirm closing the project, stating that applications still under review will be withdrawn. 2. On confirmation, the system marks the project listing as closed and removes it from student browsing. 3. The system withdraws the applications to the project that are still under review, moves each affected Student's remaining preferences up, and notifies those Students. Accepted students and open offers are kept. AF-S3: If the Project Supervisor publishes a draft. 1. The Project Supervisor selects PUBLISH on a listing under Drafts. 2. The system publishes the listing if the application cycle has not closed. AF-S6: If the Project Supervisor selects CANCEL. 1. The system discards the entered data and returns to the project listing screen.

Exceptions:	E1: A required field is missing or invalid, or a text is over its limit. 1. The system rejects the submission. 2. The system displays an error message identifying the invalid field and prompts the Project Supervisor to correct it. 3. The system remains on the project configuration page. E2: The Project Supervisor attempts to reduce the number of slots below the students who have accepted a place plus the match offers still open. 1. The system rejects the change. 2. The system displays an error message stating the minimum permitted number of slots. 3. The system remains on the project configuration page. E3: An uploaded image is not a real JPEG or PNG file, or is larger than 5 MB. 1. The system rejects the upload and states the reason. E4: The video link is not a YouTube or Vimeo link. 1. The system rejects the link and identifies the field. E5: The Project Supervisor attempts to update a closed listing, or any listing after the application closing date. 1. The system rejects the update and displays a message that the listing can no longer be modified. E6: The Project Supervisor attempts to publish after the application closing date. 1. The system rejects the publication. The listing may still be saved as a draft.

Includes:	-
Special Requirements:	The system shall validate all user input data. Topic tags and prerequisites may only be chosen from the standard topic list and the NTU course catalogue. The title is at most 150 characters, the description at most 5,000 and the research location at most 100. Images are checked by their content, not by the file name.
Assumptions:	The current matching cycle and its deadlines have been configured by an Admin.
Notes and Issues:	None

2.3.2 ReviewAndRankStudentApplications

Use Case ID:	#3-2		
Use Case Name:	ReviewAndRankStudentApplications		
Created By:	Robin Teoh	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review students who applied to their project and rank them in order of preference.

Preconditions:	1. Project Supervisor is logged in and authenticated. 2. The Project Supervisor has at least one published project listing. 3. At least one student has applied to the project. 4. The round's Application Window or Review Window is open.
Postconditions:	The Project Supervisor's applicant rankings are saved and recorded for use in the matching process.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The Project Supervisor selects a project to review. 2. The system displays all applicants sorted by CGPA from highest, with search by name or school and filters for school, year of study, minimum CGPA and whether prerequisites are met (REQ-VRANK-1). 3. The Project Supervisor selects an applicant to review. 4. The system displays the applicant's profile and transcript. 5. The Project Supervisor adds applicants to the ranking and reorders or removes them. 6. The system saves each change immediately.

Alternative Flows:	AF-S2: The Project Supervisor sorts the applicants differently. 1. The Project Supervisor chooses to sort by CGPA, AI score or name. 2. The system reorders the list. AF-S3: The Project Supervisor asks for AI scores. 1. The Project Supervisor selects "AI score". 2. The system rates each applicant on four factors (prerequisites, fit, experience and grades), scores each applicant from 0 to 100 against the project, and shows the reason and the quotes behind each score (REQ-VRANK-4). AF-S4: The Project Supervisor changes the weights. 1. The Project Supervisor opens "Weights" and sets how much each factor counts for the project (total 100, each at least 5). 2. The system saves the weights, and the Project Supervisor runs "AI score" again (REQ-VRANK-5, REQ-VRANK-8). Students never see the weights. AF-S5: The Project Supervisor uses "Rank top N". 1. The Project Supervisor types N and selects "Rank". 2. After the Project Supervisor confirms that the current order will be lost, the system replaces the ranking with the first N applicants as currently filtered and sorted (REQ-VRANK-11). AF-S6: The Project Supervisor changes the ranking again. 1. The system replaces the previous order with the new one, provided the round's Review Window has not closed.

Exceptions:	E1: The Project Supervisor attempts to change the ranking after the round's Review Window has closed. 1. The system rejects the change. 2. The system displays a message stating that the ranking deadline has closed. 3. The system remains on the applicant list page. E2: No Student has applied to the selected project. 1. The system displays a message stating that there are no applicants to rank. 2. The system disables the ranking functions for that project.
Includes:	-
Special Requirements:	Applicant transcripts shall be viewable within the system without requiring external software. Applicants left unranked are not matched to that listing. AI scores are shown only to the listing's own Project Supervisor.
Assumptions:	Students have completed their profiles before applying.
Notes and Issues:	Rankings must be saved before the deadline set by the Admin. Applicants a Project Supervisor does not rank cannot be matched to that project, and the applicants page warns about unranked applicants before the deadline.

2.3.3 ApproveMatchingResults

Use Case ID:	#3-3
Use Case Name:	ApproveMatchingResults

Created By:	Robin Teoh	**Last Updated By:**	Robin Teoh
Date Created:	26th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review students proposed to be matched to their project and decline any match during the Approval Window. Matches that are not declined are accepted automatically when the window ends.
Preconditions:	1. Project Supervisor is logged in and authenticated. 2. Matching has run automatically and the round's Approval Window is open. 3. At least one proposed match exists for the Project Supervisor's project.
Postconditions:	Each declined match is recorded with its reason, if one was given, and its slot is freed. Matches that are not declined are accepted when the Approval Window ends and published with the round's results.
Priority:	High
Frequency of Use:	Low

Flow of Events:	1. The system displays the students proposed to be matched to the Project Supervisor's project, grouped by project, while the round's Approval Window is open. 2. The Project Supervisor selects a proposed match to review. 3. The system displays the student's profile and application details. 4. If the Project Supervisor takes no action, the match is accepted when the Approval Window ends and is published with the round's results.
Alternative Flows:	AF-S5: If the Project Supervisor selects DECLINE 1. The system prompts the Project Supervisor to confirm and to give an optional reason. 2. On confirmation, the system records the decline with its reason, if one was given, and returns the slot to available. 3. Return to step 2. AF-S6: If the Project Supervisor selects CANCEL 1. The system returns to the list of proposed matches without recording a decision.
Exceptions:	E1: The Project Supervisor attempts to decline a proposed match after the Approval Window has ended or the matching results have been published. 1. The system prevents the decline. 2. The system displays a message stating that the matching results are final. 3. The system displays the proposed matches in a read-only state.
Includes:	-

Special Requirements:	None
Assumptions:	The Admin may review recorded rejections and reassign affected students from the Overrides page, before or after results are published.
Notes and Issues:	The Admin can see declines and their reasons on the round's matches page and can reassign affected students from the Overrides page. The Admin is not notified of each decline.

2.3.4 ManageSelf-ProposedProjects

Use Case ID:	#3-4		
Use Case Name:	ManageSelf-ProposedProjects		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh
Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	Project Supervisor
Description:	Allows a Project Supervisor to review research proposals pitched directly by students and either accept & register or decline them.
Preconditions:	1. The Project Supervisor is logged in. 2. At least one student has submitted a self-proposed project to this supervisor.
Postconditions:	The proposal status is updated, and the student is either registered to the supervisor or notified of rejection.

Priority:	High
Frequency of Use:	Several times during an active application cycle
Flow of Events:	1. The Project Supervisor navigates to the "Proposals" page. 2. The System displays all proposals submitted by students directed to the supervisor, with their status. 3. The Project Supervisor reviews the proposal title, writeup, research tags, the student's profile and supporting documents. 4. The Project Supervisor clicks "Accept and register". 5. The System asks the Project Supervisor to confirm, stating that the student's other applications, open offers and pending proposals will be withdrawn. 6. The Project Supervisor clicks "Accept and register" to confirm. 7. The System registers the student to a new one-slot project under the supervisor, withdraws the student's other open applications, open offers and pending proposals, notifies the Supervisors concerned, makes the student ineligible for later rounds, and notifies the student.
Alternative Flows:	AF-S4: Supervisor declines proposal. 1. The Project Supervisor clicks "Decline" and enters a reason of at least 10 characters. The System marks the proposal as declined, stores the reason, shows it to the student with the declined proposal, and notifies the student with that reason (REQ-REVPROP-3).

Exceptions:	E1: Application Cycle has closed. 1. The System blocks acceptance and displays a notice. The proposal can still be declined. E2: The student already holds another project, or has declined or missed a match offer. 1. The System blocks acceptance and explains why. The proposal can only be declined. E3: The student withdrew the proposal. 1. The System shows the proposal as Withdrawn, with no actions.
Includes:	-
Special Requirements:	Accepting a proposal shall automatically trigger an email/system notification to the student.
Assumptions:	A student can only be registered to one confirmed URECA project per cycle.
Notes and Issues:	Direct supervisor acceptance bypasses the automated matching algorithm round.

2.4 For Functional Requirement #4 (User)

2.4.1 ViewNotifications

Use Case ID:	#4-1		
Use Case Name:	ViewNotifications		
Created By:	Edwin Tan	**Last Updated By:**	Robin Teoh

Date Created:	29th August 2026	**Date Last Updated:**	1st November 2026

Actor:	User (Student, Project Supervisor, Admin)
Description:	Allows an authenticated user to view incoming and historical system notifications regarding status changes, application updates, cycle dates, and matching outcomes.
Preconditions:	The User is logged in and authenticated.
Postconditions:	Notifications are displayed, and selected unread items are updated to a read state in persistent storage.
Priority:	Medium
Frequency of Use:	High (frequently checked during active application and matching periods)
Flow of Events:	1. The User clicks on the "Notifications" item in the navigation menu. 2. The System queries and displays the list of relevant timestamped notifications (e.g., application status changes, new applicant alerts, match results, or deadline updates). 3. The User selects a notification to view its complete message details. 4. The System marks the selected notification as read and updates the unread badge counter.
Alternative Flows:	AF-S3: User navigates via notification interactive link The User clicks an embedded link inside a notification (e.g., "View Application" or "View Match Result"). The System redirects the User directly to the target record page

Exceptions:	E1: Notification service fetch timeout or server failure 1. The System displays an error message informing the user that notifications could not be retrieved.
Includes:	-
Special Requirements:	In-app notifications shall be delivered and displayed within 1 minute of the triggering event
Assumptions:	All relevant status-change triggers automatically persist notification records for affected user accounts.
Notes and Issues:	Email notifications, where applicable, are dispatched asynchronously within 15 minutes of the event.

2.4.2 SignIn

Use Case ID:	#4-2		
Use Case Name:	SignIn		
Created By:	Ian Tan	**Last Updated By:**	Robin Teoh
Date Created:	31st August 2026	**Date Last Updated:**	1st November 2026

Actor:	User (Student, Project Supervisor, Admin)
Description:	Allows a registered, verified user to securely access the URECAMatch platform using their email address and password.

Preconditions:	The User has a registered and activated account.
Postconditions:	The user is authenticated and granted access to their respective dashboard.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The User navigates to the sign-in page. 2. The System displays input fields for NTU email address and password. 3. The User enters their credentials and clicks "Sign in". 4. The System validates the credentials against the user database. 5. If valid, the System authenticates the user and redirects them to their respective dashboard.
Alternative Flows:	-
Exceptions:	E1: Invalid credentials provided. 1. The System displays an error message and prevents access. E2: Account is deactivated. 1. The System prevents sign-in and displays an error message. E3: Account is not yet verified. The System prevents sign-in and prompts the User to complete e-mail verification.
Includes:	-

Special Requirements:	Password input shall be masked, with an option to reveal it. Session shall end after 30 minutes without an authenticated request.
Assumptions:	Users have already successfully completed the registration and email verification process.
Notes and Issues:	-

2.4.3 RegisterAccount

Use Case ID:	#4-3		
Use Case Name:	RegisterAccount		
Created By:	Ian Tan	**Last Updated By:**	Ian Tan
Date Created:	31st August 2026	**Date Last Updated:**	19th October 2026

Actor:	User (Student or Project Supervisor; Admin accounts are created by an Admin)
Description:	Allows a user to create a new account using an NTU email address and password, followed by an email verification process to activate the account.
Preconditions:	None (User is not yet registered).
Postconditions:	A new user account is created in the system and activated upon successful verification.
Priority:	High

Frequency of Use:	High (at start of application cycle)
Flow of Events:	1. The User navigates to the registration page. 2. The System displays input fields for full name, NTU email address, password, and account role selection (Student or Project Supervisor). 3. The User enters details and clicks "Create account". 4. The System validates the email domain and checks for existing accounts. 5. The System sends a 6-digit verification code, valid for 10 minutes, to the provided email address. 6. The User enters the verification code. 7. The System verifies the code and activates the account.
Alternative Flows:	AF-1: User requests a new verification code if the previous one expired or was not received.
Exceptions:	E1: Invalid NTU email domain provided. E2: Email address already associated with an existing account. E3: Incorrect or expired verification code entered. E4: Password shorter than 8 characters. The System rejects the registration and displays the password rule.
Includes:	-
Special Requirements:	Registration is restricted to NTU email domains (@e.ntu.edu.sg or @ntu.edu.sg) or approved external domains. Password input shall be masked.

Assumptions:	Users have access to their provided email account to receive the verification code.
Notes and Issues:	-